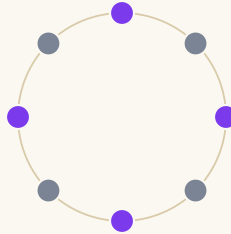


The Compass and the Ear

*What Islamic geometric ornament and the metre circles of al-Khalīl
have in common — and where they differ*



the compass



the ear

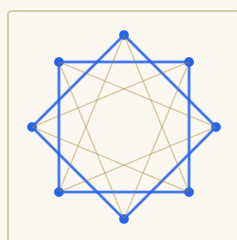
*“A mathematician, like a painter or a poet, is a maker of patterns. If his
patterns are more permanent than theirs, it is because they are made with
ideas.”*

— G. H. HARDY, *A Mathematician’s Apology* (1940)

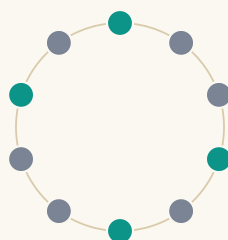
§1 Two Workshops, One Idea

Two crafts, centuries apart. In eighth-century Basra, al-Khalīl ibn Aḥmad works out that every classical Arabic metre comes from five circles of repeating sound-units — the subject of chapter I. In a workshop in Isfahan or Fez or Granada, a craftsman sets a compass on a blank panel, divides a circle into equal parts, and lays out the strapwork that will cover a wall with stars.

The two crafts never spoke. One works in sound, the other in tile and plaster. Both rest on the same premise: a small kit of parts, repeated under strict rules, produces unlimited variety. Analysed with modern mathematics, the same structures appear in each — the groups, orbits and stabilizers of chapter I, plus several that exist only in two dimensions.



the star $\{8/2\}$, framed for a wall



*the ten-unit necklace of *المختلف**

Fig. 1 — *The same kind of object in both trades. Left: the eight-pointed star $\{8/2\}$, the seed of countless Islamic panels. Right: the ten-unit circle of *المختلف* from chapter I. Each is a divided circle read by a rule.*

This chapter is a guided comparison. The route: the shared construction method — a circle that gets erased (§2); a shared vocabulary of stars, including one figure that appears identically in Qur'an manuscripts and in the metre circles (§3); the big difference — mirrors work in space but not in time, with one computed surprise (§4); strips and grids, where the eye out-scores the ear seven to one and then seventeen to one (§5); the rules each medium imposes, and the famous workaround (§6); why both arts break their own symmetry on purpose (§7); and finally a side-by-side dictionary of the whole correspondence (§8). The claim is structural, not mystical, and every number in it can be checked.

§ 1 · CONTINUED

Start with the parts lists. Both are short. Chapter I built every classical metre from four units: the light cord $\emptyset/$, the joined peg $\emptyset//$, the split peg $/\emptyset/$, and the cluster $\emptyset///$. Four bricks, sixteen metres. On the ornament side, when researchers reconstructed how the most intricate five-fold patterns could have been laid out, they arrived at a kit of just five tiles — the girih tiles — all with the same edge length, each carrying the same angled guide-lines. Five shapes, and enough patterns for walls from Baghdad to Bukhara.

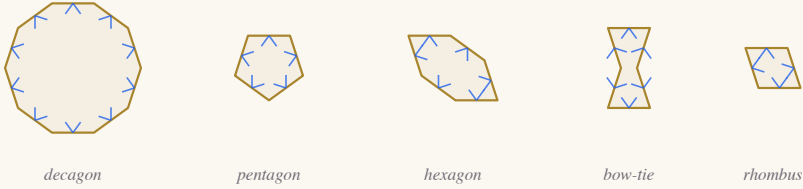


Fig. 2 — The girih kit: decagon, pentagon, elongated hexagon, bow-tie, rhombus. All edges are the same length; the blue marks show where the strap lines cross each edge. Below: the prosodist's kit — the four units that generate all sixteen metres.



A fair objection: every craft has parts, so a short parts list proves nothing by itself. The real link is what both traditions do with their kits. Neither arranges parts freely. Both generate their designs from a circle plus a counting rule — which means the space of possible designs can be listed completely. Every possible metre; every possible star from a given division. This turns a repertoire into a catalogue, and catalogues are what mathematics can analyse. That is the link the rest of the chapter follows.

§2 Start with a Circle — Then Hide It

Here is how a geometric panel is actually made. Step one: draw a circle. Step two: divide it into n equal parts. Step three: connect the division points by a counting rule — every second point, every third. Step four: erase the construction. The finished wall shows the star; the circle that produced it is gone. Nearly every Islamic geometric design, however complex it looks, starts exactly this way.

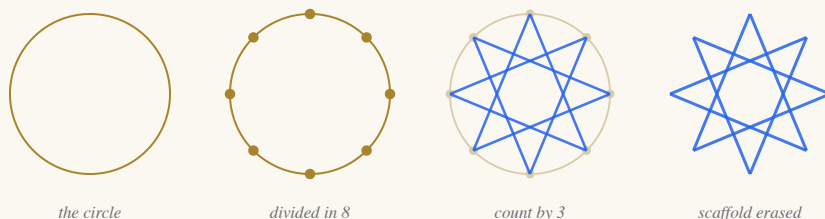


Fig. 3 — An eight-pointed star in four steps: draw a circle; divide it into eight; connect every third point ($\{8/3\}$); erase the scaffolding.

Now compare the prosodist. Chapter I showed that a metre is built the same way: a circle of units, a chosen starting point, the units read off around the loop. And then the circle disappears — no poem ever shows its *dā'ira*. Listeners get a line of verse; the wheel that generated it appears only in the technical manuals.

So both arts have a two-layer structure. The public layer is the artifact: the wall, the poem. The private layer is the generating circle. The detail that matters: neither tradition discarded its private layer. Both wrote it down, in a specialist literature of diagrams — which means both understood that the circle, not the surface, is where the design really lives. The next page shows the paper trail.

§ 2 · CONTINUED

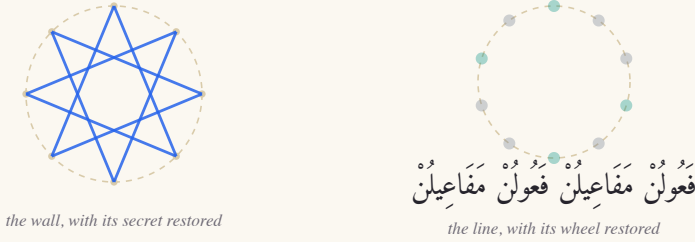


Fig. 4 — The hidden layer, restored. Left: a finished star with its construction circle drawn back in. Right: a line of الطويل with its unit-circle drawn behind it — the *dā'ira* no listener ever sees.

The paper trail

The link between geometers and craftsmen is documented, not guessed. In tenth-century Baghdad, the mathematician **Abū al-Wafā' al-Būzjānī** wrote a practical manual — *On the Geometric Constructions Necessary for the Artisan* — and mentions working sessions where geometers and craftsmen solved problems together. The **Topkapı scroll** (studied by Gülru Necipoğlu, 1995) is about thirty metres of construction drawings: a working pattern book that preserves exactly the scaffolding the finished walls hide. The prosodists' manuals, with their circle diagrams of the sixteen metres, are the same kind of document for poetry.

This two-layer picture also clarifies what “abstract art” means here. The structure comes first; any particular wall or poem is one rendering of it. Change the tiles' colours and it is still the same pattern; change the poem's words and it is still الطويل. The circle is the master copy, and both traditions gave the master copy its own literature.

§3 A Shared Alphabet of Stars

If the two systems share one visual, it is the star polygon. The recipe: put n points evenly on a circle, then connect every k -th point. The result is written $\{n/k\}$. Chapter I met these stars as the drawn form of a symmetry — connect each unit of a metre circle to the unit its rotation-symmetry carries it to, and you get the star $\{n/p\}$. Islamic ornament, free to choose any n and k , made the same figures its favourite vocabulary:

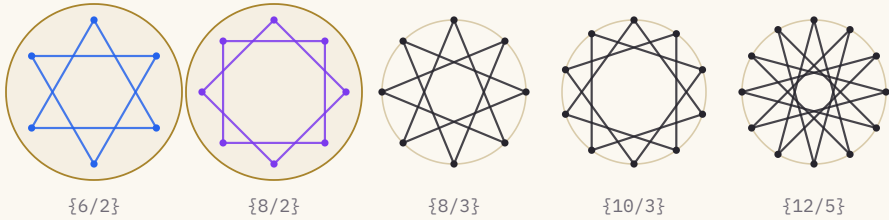


Fig. 5 — Five common star polygons. The two on gold backgrounds — $\{6/2\}$ and $\{8/2\}$ — are exactly the symmetry stars of circles 2 and 5 from chapter I. The other three, $\{8/3\}$, $\{10/3\}$, $\{12/5\}$, are favourites of the ornamental tradition that no metre circle happens to produce.

One arithmetic fact controls how these stars look: the greatest common divisor $\gcd(n, k)$. If $\gcd(n, k) = 1$, the star is a single unbroken path — you can draw $\{8/3\}$ without lifting the pen. If $\gcd(n, k) = d > 1$, the star splits into d separate interlocking polygons: $\{6/2\}$ is two triangles, $\{8/2\}$ is two squares, $\{9/3\}$ is three triangles. The interlace so typical of Islamic design — separate closed bands woven together — is this divisor arithmetic made visible. (Group theory calls the d bands the cosets of a subgroup; the picture is all you need to keep.)

§ 3 · CONTINUED

Now the best single example in the chapter.

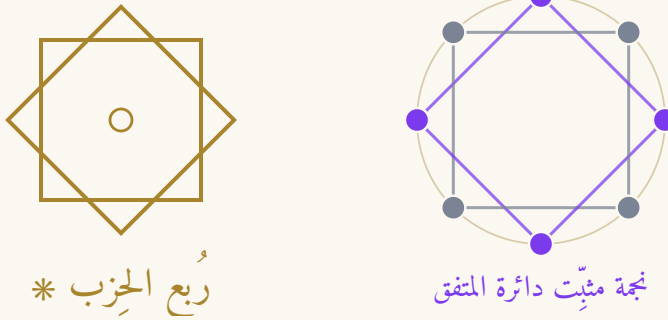


Fig. 6 — Left: the *rub' al-hizb* * — two overlapping squares, $\{8/2\}$ — the mark that indicates the quarter-divisions of the *hizb* in Qur'an manuscripts. Right: the symmetry star of circle 5 (المتفق), drawn by the circle's own C_4 rotation on its own eight units, exactly as in chapter I.

These are the same figure. The mark that has divided the text of the Qur'an for a thousand years is exactly the star that circle 5's symmetry draws on its eight units. And nobody copied anyone. One practice divides a sacred text into quarters and eighths for recitation; the other divides a circle of sounds into readings. Both perform the same underlying operation — a cycle of eight, stepped through two at a time — so both produce the same shape. That is how a shared mathematics shows itself: not as resemblance, but as identical objects reached from different directions.

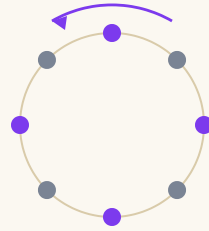
The hexagram gives a second example. As ornament it is the *khātam* — the Seal — one of the oldest devices in the region's art. As mathematics it is $\{6/2\}$. And in chapter I it appeared as the symmetry star of circle 2, where its two triangles even carry meaning: one triangle connects the three pegs, the other the three clusters. Same figure, three jobs.

§4 Mirrors: Where the Two Arts Part Ways

Here the two arts diverge, and the difference is the most informative thing in the chapter. Take an eight-fold rosette. You can rotate it an eighth of a turn and it looks unchanged — eight rotations in all. But you can also *flip* it, across eight mirror lines. Rotations plus reflections make sixteen symmetries: the dihedral group D_8 . Ornament gets all sixteen, because ornament lives in space, and space has no preferred direction.



D_8 — eight turns, eight mirrors



C_8 — eight turns, no mirror: time's arrow

Fig. 7 — Left: an eight-fold rosette with its eight mirror lines; with the eight rotations, its symmetry group is D_8 — sixteen moves. Right: the circle of *التنق* gets rotations only — C_8 at most — because a rhythm plays in one direction.

Now try to flip a metre. A reflected rhythm is the rhythm played backwards — and you cannot listen backwards. Sound arrives in time order, and time runs one way. So of the sixteen symmetries the eye enjoys on a rosette, the ear can use at most the eight rotations. This is why every group in chapter I was cyclic (C_n) and never dihedral (D_n). It is not a modelling choice. It is a fact about the medium.

Islamic art uses both options. Most patterns have full mirror symmetry; some deliberately do not: pin-wheel motifs in brickwork rotate without reflecting — their group is a bare C_n . And when calligraphers do mirror text, in the style called *muthannā* (a phrase facing its own reflection), they mirror the *written image* of the words, never the sound. Writing lives in space, so it can be flipped. Speech lives in time, so it cannot.

§ 4 · CONTINUED

This raises a question neither tradition could have asked, but we can: *do al-Khalil's circles have mirror symmetry they can't use?* Take a metre circle, write it out as its ring of struck and quiet letters, and lay it flat like ornament — where flipping is allowed. Is the flipped ring the same ring?

The check is mechanical: reverse the letter sequence, then test whether some rotation brings it back to the original. Run that on all five circles and here is the result:

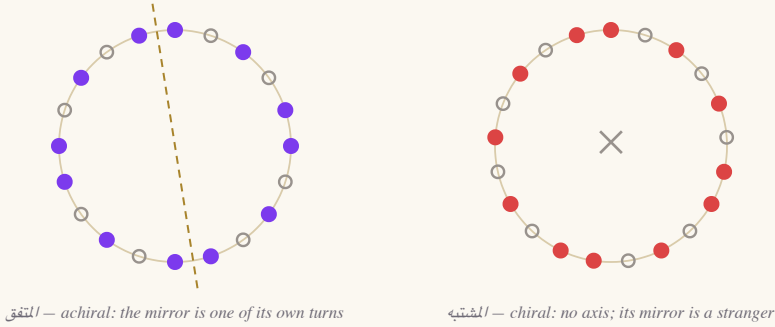


Fig. 8 — Letter-rings of two circles (● = struck letter, ○ = quiet letter). Left: circle 5 (المتفق). Reflecting across the dashed axis maps the ring onto itself — it is achiral. Four of the five circles pass this test. Right: circle 4 (المشتبه). No such axis exists — the flipped ring matches no rotation of the original. It is chiral, like a left hand.

Four of the five circles are achiral: laid out in space, each equals its own mirror image after some rotation. Their full spatial symmetry is dihedral — larger than anything the ear can perform. The metres carry mirror symmetries that can only be seen, never heard. It took ornament's point of view to find them.

And the one exception is circle 4, المشتبه — the same circle chapter I found had no rotational symmetry at all (the one-peg story). It turns out to have no mirror either. It is the only chiral necklace in the system: asymmetric under every test, in every geometry. The tradition called it “the confusable.” Every measurement we have applied keeps agreeing.

§5 Patterns on a Strip, Patterns on a Plane

Between a single rosette (pattern around a point) and a full wall (pattern across a plane) sits the border: pattern along a strip. Strip patterns have been classified completely — there are exactly **seven** possible symmetry types, the frieze groups. Islamic ornament uses all seven, in manuscript frames, cornice bands, and woven trims.

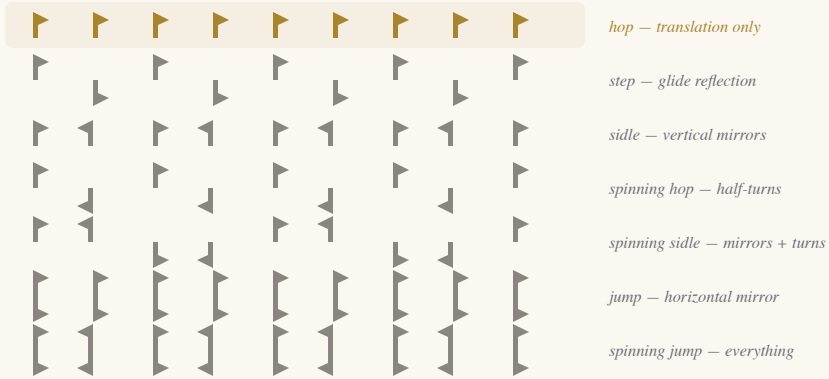


Fig. 9 — The seven frieze types, all built from one asymmetric motif. Row 1 repeats by translation only; the others add glides, vertical mirrors, half-turns, and combinations. A line of verse is also a strip pattern — but it can only be row 1 (gold), because every other row flips something.

A line of metred verse is a strip pattern in sound. The foot repeats — **فَعُولُنْ مَفَاعِيلُنْ فَعُولُنْ مَفَاعِيلُنْ** — and repetition is translation: row 1. The other six rows all involve a reflection or a half-turn: playing the strip backwards, or upside down. For rhythm, backwards is unperformable and upside down is not even defined. So of the seven-row catalogue, the ear's border art occupies exactly one row.

Music does use the forbidden rows — a crab canon plays a melody against its own reversal. But note where that happens: on the page. Notation lays time out as space, and space can be read both ways. It is the musician's version of muthannā — a mirror applied to the *picture* of time, since it cannot be applied to time itself.

§ 5 · CONTINUED

Now go up one more dimension. A pattern that repeats in two directions — a wallpaper, a pavement, the tiled lower walls of a madrasa — can have exactly **seventeen** symmetry types: the wallpaper groups, proved complete by Fedorov in 1891 and independently by Pólya in 1924. The famous claim that the Alhambra contains all seventeen is contested: careful counts range from eleven to seventeen, depending on how colour and restoration are treated. But the exact number matters less than the fact behind it. Craftsmen explored a large share of a seventeen-entry catalogue, centuries before anyone knew the catalogue existed — or that it was finite.

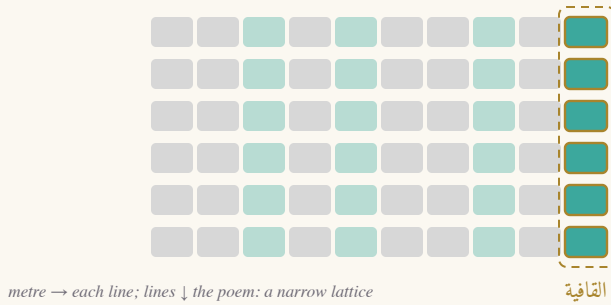


Fig. 10 — Poetry’s version of a two-dimensional pattern. Each row of a *qasida* repeats the same metre (horizontal repetition); the rows stack (vertical repetition); and one column is fixed — the *qāfiya*, the rhyme every line ends on (outlined in gold).

Does poetry have anything two-dimensional? Almost. A classical *qasida* is a stack of lines in one metre, and every line ends on the same rhyme, sometimes for a hundred lines. As pattern, that is a genuine lattice: repeat across (the foot), repeat down (the verse), with one marked column. But it is a lattice with almost no symmetry beyond the two translations — mirrors and turns are still ruled out, for the reasons of §4. Keep the scoreboard in mind: on a point, the eye gets D_n and the ear C_n ; on a strip, seven types against one; on the plane, seventeen against a bare grid. Every added dimension widens the eye’s lead, because every added dimension brings more ways to flip — and flipping is the one move sound can never make.

§6 Every Medium Has Rules

Chapter I found a law limiting a metre circle’s symmetry: the symmetry’s size must divide the number of beads (Lagrange’s theorem — divisors only, nothing in between). The plane has its own law, and it is more surprising. A pattern that repeats in two directions can contain rotation centres of order 1, 2, 3, 4, and 6 — *and nothing else*. No five-fold centre. No seven-fold. Ever, in any periodic pattern. This is called the crystallographic restriction.

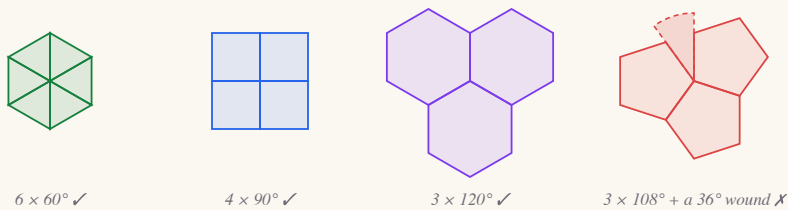


Fig. 11 — Why there is no five-fold wallpaper. Six triangles close neatly around a point; so do four squares and three hexagons. Three regular pentagons leave a 36° gap, and a fourth would overlap. What cannot close around a point cannot repeat across a plane.

WORLD	ITS OBJECTS	ITS LAW OF SYMMETRY
the necklace (time’s loop)	metres	stabilizer order divides n — any n welcome
the plane (space’s field)	tilings	rotation orders 1, 2, 3, 4, 6 only

Fig. 12 — The two rulebooks, side by side.

Notice the trade-off between the two rulebooks. The circle of beads is generous about size — al-Khalil’s circles use 6, 8, 9, and 10 beads without trouble — but strict about which symmetries fit a given size (divisors only). The plane is generous about size and strict about kind: whatever the pattern, five-fold symmetry is banned outright. That ban mattered, because Islamic artists loved five- and ten-pointed stars. The next page shows the two ways they got them anyway.

§ 6 · CONTINUED

The first workaround is straightforward: put the ten-pointed star inside a local medallion and let the wall's actual repetition obey the rules. The star is local; the lattice stays legal. Hundreds of buildings use this compromise.

The second workaround is the well-known one. In the girih tradition of Iran, large patterns were built by subdividing big tiles into smaller copies of the same shapes. Subdivision does not need a repeating cell, so it quietly steps around the restriction. At the Darb-i Imam shrine in Isfahan (1453), the result comes remarkably close to a *quasicrystalline* pattern: five-fold order that never repeats — the structure Roger Penrose discovered mathematically in the 1970s.

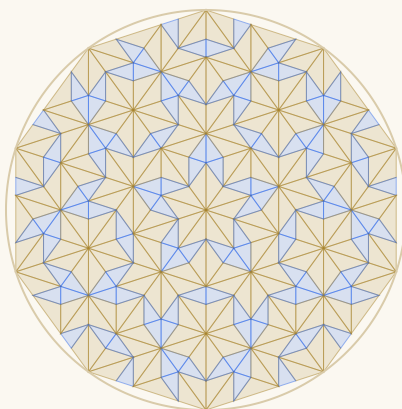


Fig. 13 — A five-fold pattern with no repeating unit, generated by golden-ratio subdivision (shown here through the Penrose rhombi's triangle halves). Patterns in this family are perfectly ordered but never periodic. The Darb-i Imam girih approximates one.

The claims and the caveats, in order. Peter Lu and Paul Steinhardt published the quasicrystal reading of Darb-i Imam in *Science* in 2007. Emil Makovsky had argued a related case for the 1197 Maragha tomb tower years earlier. Peter Cromwell and others have pushed back, favouring traditional construction methods with no quasi-periodic intent. And the standard qualifier is “nearly perfect” — the physical pattern has defects. What nobody disputes is the mathematics: patterns of this family are real, they achieve five-fold order by giving up periodicity rather than law, and girih craftsmen were building convincing approximations of them five centuries before Penrose. The restriction said five-fold order cannot *repeat*. The craftsmen found the sense in which order does not need to.

§7 Why Perfect Symmetry Is Boring

In a tiled interior the eye does not settle on the perfectly repeating field; that reads as background. It goes to where the repetition breaks — the miḥrāb, the central medallion, the calligraphy band. Designers use symmetry as *ground* and controlled symmetry-breaking as *figure*. The repetition sets an expectation and the break spends it.

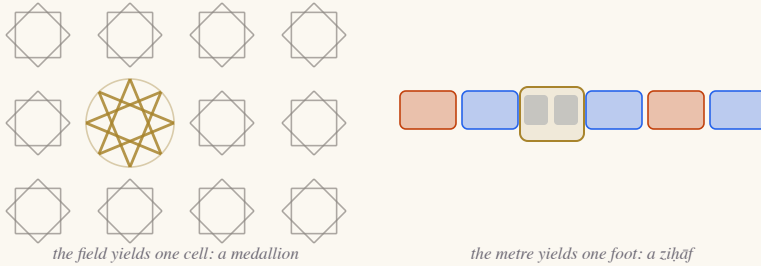


Fig. 14 — The same design move in both arts. Left: a periodic star field with one cell opened into a larger medallion — the break is where the eye goes. Right: a metric line with one foot altered by a *zihāf* (gold) — the variation is what the ear notices.

Arabic prosody works exactly the same way. The metre is the ground: fixed, predictable, the listener’s baseline. The *zihāf* system — the licensed foot-variations mapped in this project’s confusion-map view — is the figure: a syllable quieted here, a foot lightened there, always inside strict limits. A poem with zero variation sounds mechanical; one with too much loses the metre. Good poets, like good tilers, worked the boundary between the two.

Chapter I said the same thing with numbers. Its “inverse law” showed that the most symmetric circle produces the fewest metres: symmetry is sameness, and sameness collapses options. Anything you actually notice — a metre, a medallion, a memorable line — exists because some symmetry was deliberately given up. Both arts learned this in practice, long before there was a theory to say it in.

§8 The Two Systems, Side by Side

The full correspondence, one table. Left column: ornament, which lives in space. Right column: metre, which lives in time. Each row pairs a concept, not a mood.

ORNAMENT (SPACE)	METRE (TIME)
motif — the repeated device	the foot (تَفْعِيلَة)
construction circle, divided in n	the dāʿira of n units
unit cell / fundamental region	the period p — chapter I’s arc
rosette symmetry D_n (with mirrors)	stabilizer C_p (mirrors unheard)
star polygon $\{n/k\}$, interlaced cosets	stabilizer star $\{n/p\}$ on the necklace
frieze — 7 lawful strips	the verse line — 1 lawful strip
wallpaper — 17 lawful planes	the qaṣīda’s narrow lattice
crystallographic restriction (2,3,4,6)	Lagrange’s divisor ladder
catalogue explored by craft (Alhambra)	catalogue derived by ear (21 rotations)
giriḥ: order beyond periodicity	— (time’s line is too narrow)
erased scaffold; the Topkapı scroll	silent circle; the prosodists’ diagrams
ground/figure: field and medallion	metre and zihāf

Fig. 15 — The correspondence, concept for concept. One row has no right-hand entry — and that blank is informative too.

Two mismatches are worth naming, because the table is honest about them. Ornament has nothing like the muḥmal metres — patterns that are lawful but that a canon declined to use — because walls have no canon of sixteen. And metre has nothing like the quasicrystal: a one-dimensional loop simply has no room for that kind of order. So the picture is not “two arts, secretly identical.” It is: a large shared core, plus one private wonder on each side. That is what “one mathematics, two instruments” actually means.

§9 Two Instruments, One Method

A compass makes equal things: every point it draws sits at the same distance from the centre. An ear detects them: two durations, two stresses, alike or not. One instrument constructs symmetry, the other verifies it. Give both the same task — repeatable pattern under strict rules — and each maps out whatever mathematics its medium allows.

The compass had the larger territory: seventeen wallpaper groups, free mirrors, and the quasi-periodic loophole at the edge. The ear had less — cyclic groups only, one frieze row, a thin lattice. But al-Khalil did something the artisans never needed to: he worked his catalogue out completely and showed it was closed. Twenty-one rotations, sixteen in use, five named or noted. The artisans explored their catalogue. Al-Khalil mapped his.

Both fit their own culture's theory of beauty, too. Ibn al-Haytham's *Book of Optics* locates beauty not in objects but in relations between parts — proportion, order, composition. A tiling and a metre are beautiful in precisely that register. The civilization had a theory of why lawful pattern pleases, and built two major arts that run on it.

What this chapter does not claim

No secret geometry of the poets, no numerology of the tilers, no hidden transmission. Craftsmen thought in workshop recipes; prosodists thought in sounds and licenses. The claim is structural, which makes it checkable: examined with modern tools, both repertoires turn out to be organized by the same mathematical objects — group actions, orbits, stabilizers, symmetry laws. Each craft obeyed its medium, and the media obey mathematics. The convergence is the finding.

§10 The Names Came Later

Chapter I ended with a date: the 1830s, when Galois named the group — a thousand years after al-Khalil used one. This chapter adds two more dates. Fedorov proved the seventeen wallpaper groups in 1891; the Alhambra’s builders had explored much of that catalogue by the 1300s. Penrose found aperiodic five-fold order in the 1970s; the Darb-i Imam girih approximated it in 1453.

Three times, the same sequence: practice first, mathematics later. When a craft commits over generations to one medium and strict rules, the structure of that medium shows up in the work, whether or not anyone has words for it. The mathematics in this chapter was not applied to these two arts. It was recovered from them.

Chapter takeaways

- Both arts generate their works from a divided circle that the finished piece hides (§2).
- Star polygons $\{n/k\}$ are shared vocabulary — and $\odot$ is literally circle 5’s symmetry star (§3).
- Space allows mirrors; time does not. Ornament gets D_n , metre gets C_n (§4).
- Computed here: four of the five metre circles are secretly mirror-symmetric; المشتبہ alone is chiral (§4).
- Strips: 7 symmetry types for the eye, 1 for the ear. Planes: 17 against a bare lattice (§5).
- Each medium has its own law — divisors for bead-circles, $\{1,2,3,4,6\}$ for the plane — and girih artists bypassed the plane’s law by dropping periodicity, not order (§6).
- In both arts, what you notice is where the symmetry breaks (§7).

The prosodist turning a circle of sounds and the craftsman turning a compass were working on the same object — the group — from opposite sides of the divide between time and space. Neither could see the other’s side. Twelve centuries later the two halves fit.

For the Hands and the Ears

Draw one

Take a real compass. Draw a circle, keep the same radius, and walk it around the circumference: it fits exactly six times. That gives you $\{6/2\}$ and $\{6/3\}$ for free — join alternate points for the Seal, join opposite points for the wheel. For the eight-star $\otimes$, bisect twice instead. This is the opening move of every pattern scroll, and the same move al-Khalil's circle diagrams make in sound.

Hear one

In the Interactive Arud Explorer (arudi.midadalfikr.com), open the Mathematics view and pick دائرة المتفق: its C_4 symmetry draws the two squares of Fig. 6 on the circle itself. Then open the Rhythm Clock on the same circle and press play. The four-fold return of the low drum is the $\otimes$ as a sound: the same object, once through the eye and once through the ear.

Break one, lawfully

In the Confusion Map view, play الكامل and switch on الإضمار while it sounds: one drum stroke falls silent. That is $\$7$ in miniature — a lawful break in a regular pattern, the audio version of the one medallion cell in a repeating field.

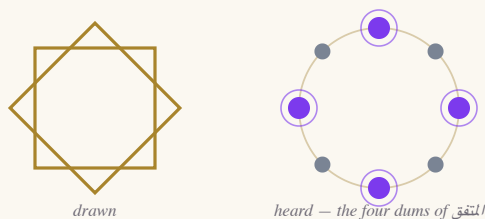


Fig. 16 — One object, two senses: the eight-star as compass-work, and the circle of المتفق with its four drum-beats (the pegs) ringed.

Glossary

gīriḥ گره

“knot”: the Persian strapwork style; also the five-tile kit that generates its five-fold patterns (§1, §6).

zellij / zillīj زليج

cut-tile mosaic of the Maghrib and al-Andalus.

rosette

a point-symmetric star device; symmetry C_n or D_n (§4).

dihedral group D_n

the n rotations *and* n reflections of an n -fold figure — twice the cyclic group C_n of chapter I (§4).

chiral / achiral

distinct from its mirror image / equal to it. Four necklaces are achiral; المشتبه alone is chiral (§4).

frieze groups

the seven symmetry types of strip patterns; sound can perform exactly one (§5).

wallpaper groups

the seventeen symmetry types of doubly periodic plane patterns (Fedorov 1891, Pólya 1924) (§5).

crystallographic restriction

periodic plane patterns admit rotation orders 1, 2, 3, 4, 6 only — the plane’s counterpart of Lagrange’s ladder (§6).

quasiperiodic

perfectly ordered, never repeating; Penrose tilings, and (nearly) the Darb-i Imam girih (§6).

khātam خاتم

the Seal: hexagram {6/2} — the stabilizer star of (3§) المونلف.

rub‘ al-ḥizb رُبّ الحِزب

Qur’anic division mark: two squares, {8/2} — the stabilizer star of (3§) المتفق.

muthannā مثنّى

mirror calligraphy: reflecting the image of words, since their sound cannot be reflected (§4).

muqarnas, miḥrāb, dado

vault ornament of tiered cells; prayer niche; the lower wall zone that carries the periodic field (§7).

qāfiya قافية

the monorhyme pinning the qaṣīda’s end-column (§5).

ziḥāf زحاف

licensed metrical variation — the figure against the metre’s ground (§7; anatomized in the companion’s confusion map).

Sources & Notes

ON THE GEOMETRY OF ISLAMIC ORNAMENT. Abū al-Wafā' al-Būzjānī, *On the Geometric Constructions Necessary for the Artisan* (10th c.); G. Necipoğlu, *The Topkapı Scroll: Geometry and Ornament in Islamic Architecture* (1995); J. Bonner, *Islamic Geometric Patterns* (2017); S. J. Abas & A. S. Salman, *Symmetries of Islamic Geometrical Patterns* (1995); B. Grünbaum & G. C. Shephard, *Tilings and Patterns* (1987).

ON THE ALHAMBRA CENSUS. E. Müller's 1944 thesis counted eleven wallpaper groups; later studies argue for thirteen up to all seventeen. The dispute concerns rules of evidence (colour, interlace, restoration), not the mathematics.

ON GIRIH AND QUASIPERIODICITY. P. J. Lu & P. J. Steinhardt, "Decagonal and Quasi-crystalline Tilings in Medieval Islamic Architecture," *Science* 315 (2007); E. Makovicky on the Maragha tower, in *Fivefold Symmetry* (1992); P. R. Cromwell's critical re-examination, *Math. Intelligencer* (2009). The chapter reports the finding with its debates attached.

ON PATTERN, BEAUTY, AND MATHEMATICS. G. H. Hardy, *A Mathematician's Apology* (1940); H. Weyl, *Symmetry* (1952); Ibn al-Haytham, *Kitāb al-Manāẓir*, Bk II, on beauty as a relation of parts. Fedorov (1891) and Pólya (1924) for the seventeen; the frieze seven are classical.

COLOPHON. Companion to "The Turn That Changes Nothing"; same types (EB Garamond, Cormorant Garamond, Amiri, IBM Plex Mono), same 6×9 leaf. All prosodic claims are computed against the sequence data of the Interactive Arud Explorer: the stabilizer stars {10/5}, {6/2}, {9/3}, {8/2}; and the chirality census of §4 — the letter-necklaces of المتفق, المجتنب, المختلف, المؤتلف each equal their own reversal up to rotation (mirror offsets 7, 2, 2, 2 respectively), while المشتبه alone does not. Star figures are drawn by the counting rule itself; the five-fold patch of Fig. 13 by golden-ratio triangle subdivision. Prosodic symbols keep classical order: 0// reads strike–strike–rest, from the right.